

Database Design Document

DD is the process of producing a detailed data model of a database. This logical data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a Data Definition Language, which can then be used to create a database. A fully attributed data model contains detailed attributes for each entity.

The term database design can be used to describe many different parts of the design of an overall database system. Principally, and most correctly, it can be thought of as the logical design of the **base data structures** used to store the data. *In the relational model these are the **tables and views**.* In an object database the entities and relationships map directly to object classes and named relationships. However, the term database design could also be used to **apply to the overall process of designing**, not just the base data structures, but also the **forms and queries used as part of the overall database application within the database management system (DBMS).**

The process of doing database design generally consists of a number of steps which will be carried out by the database designer. Usually, the designer must:

- **Determine the relationships between the different data elements.**
- **Superimpose a logical structure upon the data on the basis of these relationships.**

The design process consists of the following steps:

1. **Determine the purpose of your database** - This helps prepare you for the remaining steps.
2. **Find and organize the information required** - Gather all of the types of information you might want to record in the database, such as product name and order number.
3. **Divide the information into tables** - Divide your information items into major entities or subjects, such as Products or Orders. Each subject then becomes a table.
4. **Turn information items into columns** - Decide what information you want to store in each table. Each item becomes a field, and is displayed as a column in the table. For example, an Employees table might include fields such as Last Name and Hire Date.
5. **Specify primary keys** - Choose each table's primary key. The primary key is a column that is used to uniquely identify each row. An example might be Product ID or Order ID.
6. **Set up the table relationships** - Look at each table and decide how the data in one table is related to the data in other tables. Add fields to tables or create new tables to clarify the relationships, as necessary.
7. **Refine your design** - Analyze your design for errors. Create the tables and add a few records of sample data. See if you can get the results you want from your tables. Make adjustments to the design, as needed.
8. **Apply the normalization rules** - Apply the data normalization rules to see if your tables are structured correctly. Make adjustments to the tables.[Database normalization is the process of structuring a relational database in accordance with a series of socalled normal forms in order to reduce data redundancy and improve data integrity].

Normalization entails organizing the columns (attributes) and tables (relations) of a database to ensure that their dependencies are properly enforced by database integrity constraints

When an attempt is made to modify (update, insert into, or delete from) a relation, the following undesirable side-effects may arise in relations that have not been sufficiently normalized:

- **Update anomaly.** The same information can be expressed on multiple rows; therefore updates to the relation may result in logical inconsistencies.

Employees' Skills

Employee ID	Employee Address	Skill
426	87 Sycamore Grove	Typing
426	87 Sycamore Grove	Shorthand
519	94 Chestnut Street	Public Speaking
519	96 Walnut Avenue	Carpentry

- **Insertion anomaly.** There are circumstances in which certain facts cannot be recorded at all.

Faculty and Their Courses

Faculty ID	Faculty Name	Faculty Hire Date	Course Code
389	Dr. Giddens	10-Feb-1985	ENG-206
407	Dr. Saperstein	19-Apr-1999	CMP-101
407	Dr. Saperstein	19-Apr-1999	CMP-201

424	Dr. Newsome	29-Mar-2007	?
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- **Deletion anomaly.** Under certain circumstances, deletion of data representing certain facts necessitates deletion of data representing completely different facts.

Faculty and Their Courses

Faculty ID	Faculty Name	Faculty Hire Date	Course Code
389	Dr. Giddens	10-Feb-1985	ENG-206
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407	Dr. Saperstein	19-Apr-1999	CMP-201

DELETE

1. GENERAL INFORMATION 1.1

Introduction

Brief write-up about database design

1.2 Purpose

Describe the purpose of the Database Specifications.

1.3 System Overview

Provide a brief system overview description as a point of reference for the remainder of the document. In addition, include the following:

1.4 Acronyms and Abbreviations

Provide a list of the acronyms and abbreviations used in **this document** (DD) and the meaning of each.

2. DATABASE IDENTIFICATION AND DESCRIPTION

2.1 Database Identification

Identify the **names or labels by which the database may be uniquely identified**

2.2 Systems Using the Database

Identify the systems that will use the database.

2.3 Database Table Structure

Table Structures *For each table, a good technical specification includes a list of the fields with the at least the following information for **every** field:*

- **field name:** a brief name which is easy to remember (e.g. "artist_name")
- **field description:** a description of the information (e.g. "the name of the artist")
- **field data type:** character, numeric, logical, date, or other; specify any specific data type such as a URL or an image. For example, the artist's name would be a character field.
- **field length:** This applies to character and numeric fields. Numeric fields can be described with a specific length for integers (e.g. "8") or with the number of places represented past the decimal (e.g. "6,2" for a number such as 999.99). Character field lengths are specified with an integer, e.g. the artist_name field could be set with a length of 30.
- **remarks:** Be sure to list any important validation rules here. For example, if the field contains the room number in a hotel reservation system, be sure that the room number applies to an actual room. Indicate here whether a field is a primary key (and if it is a foreign key, if you know that yet). Also, specify here if a field should be checked against another table (e.g. the artist's name should be validated against the table containing a list of artists on file).

There are many additional things to consider, depending on the nature of the database system that you are designing. But this is a start !!

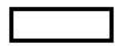
Artist Table: This table contains a list of the artists (painters and sculptors) represented by this gallery. Data fields include:

<i>field name</i>	<i>field description</i>	<i>data type</i>	<i>field length</i>	<i>remarks</i>
artist_name	the full name of the artist	character	30	Stored as "last, first" e.g. "Pollack, Jackson". This field is the primary key for this table. This field is a foreign key for the table of paintings.
artist_school	the school or style this artist is associated with primarily	character	15	This field should be checked against a list of values e.g. <i>abstract, Baroque, Impressionist</i> , etc.
artist_born	year or approximate year of the artist's birth	character	4	Use century or mid-century approximations where necessary. Be sure this date does not exceed the current year.
artist_death	year or approximate year of the artist's death	character	4	Use century or mid-century approximations where necessary. Be sure this date does not exceed the current year.
artist_nationality	the nationality this artist is associated with primarily	character	15	This field should be checked against a list of values e.g. <i>American, Mexican, Flemish</i> , etc.

3. ER DIAGRAM

Database design also includes ER(Entity-relationship model) diagrams. An ER diagram is a diagram that helps to design databases in an efficient way. **Attributes in ER diagrams are usually modeled as an oval with the name of the attribute, linked to the entity or relationship that contains the attribute.**

Within the relational model the final step can generally be broken down into two further steps, that of determining the grouping of information within the system, generally determining what are the basic objects about which information is being stored, and then determining the relationships between these groups of information, or objects. This step is not necessary with an Object database.



Represents Entity



Represents Attribute



Represents Relationship



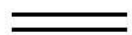
Links Attribute(s) to entity set(s) or
Entity set(s) to Relationship set(s)



Represents Multivalued Attributes



Represents Derived Attributes



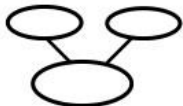
Represents Total Participation of Entity



Represents Weak Entity



Represents Weak Relationships

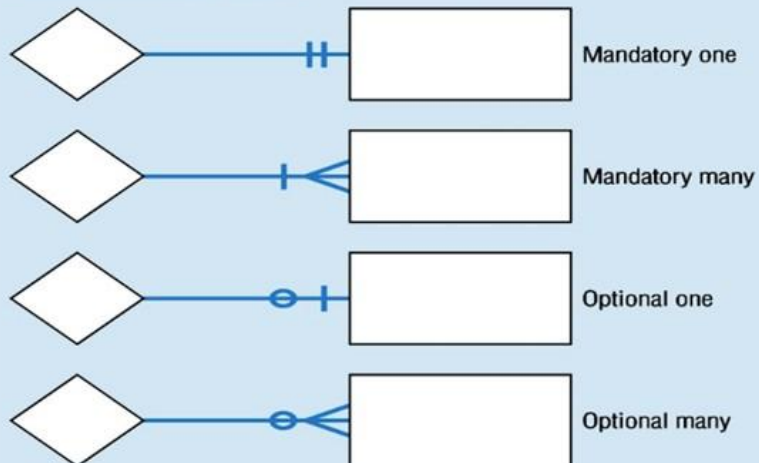


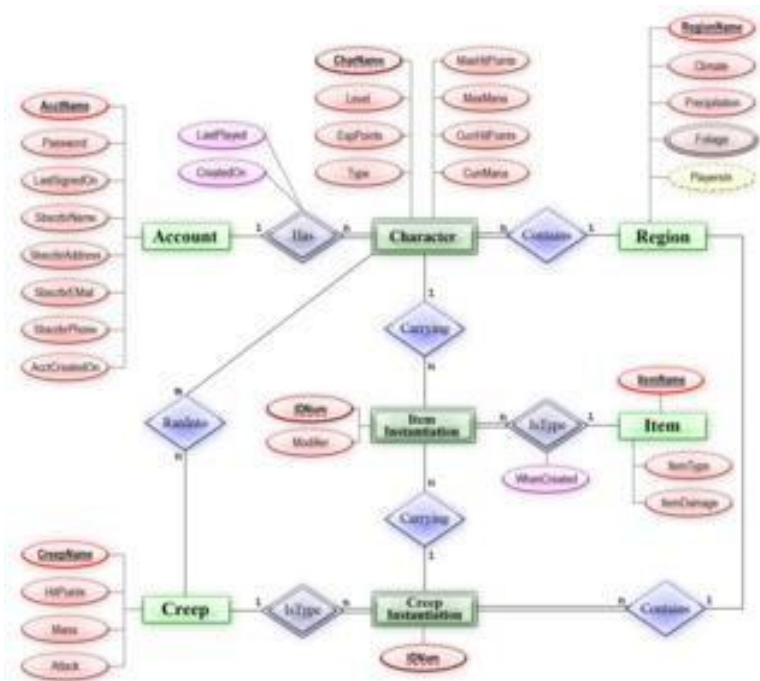
Represents Composite Attributes



Represents Key Attributes / Single Valued
Attributes

Relationship cardinality





<https://www.guru99.com/er-diagram-tutorial-dbms.html> <https://beginnersbook.com/2015/04/e-r-model-in-dbms/>